2.2.4 ▶ Carnot's Theorem

Carnot's theorem is a direct result of the second law of thermodynamics, and is an important result for engine design. It states that the most efficient heat engine that runs between two heat reservoirs is a perfectly reversible engine. The corollary of this statement is that all reversible engines running between two reservoirs of different temperature have the same efficiency.

We start the proof by considering a reversible heat engine R with efficiency η_R and an arbitrary heat engine E with efficiency η_E operating between a hot and cold reservoir.

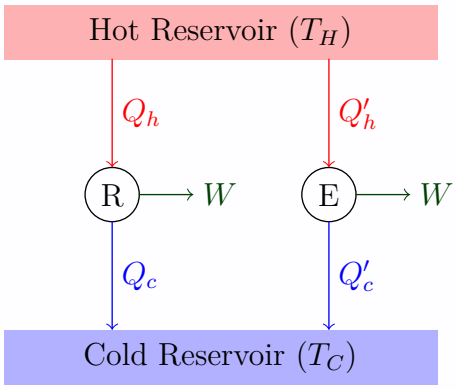

Fig. 2.3: A reversible and irreversible engine running between a hot and cold reservoir.

Both engines output work W but R takes in Q_h and outputs Q_c to T_C while E takes in Q'_h and outputs Q'_c to T_C. For the sake of the proof, assume that $\eta_E > \eta_R$, i.e.

$$\frac{W}{Q'_h} > \frac{W}{Q_h}$$

$$Q_h > Q'_h. \tag{2.31}$$

Now, consider running R in reverse using the work done by E, so that R extracts heat Q_c from T_C and deposits heat Q_h into T_H.

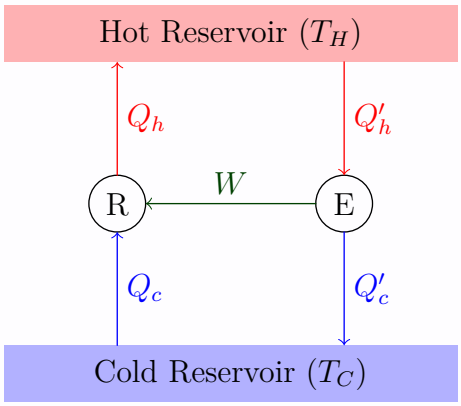

Fig. 2.4: A modified setup.

Analyzing the entire system at this state, the two engines extract heat $Q_c - Q'_c > 0$[7] from

[7] $Q_h = W + Q_c \Rightarrow$ e.q. 1.31 $\Rightarrow Q_c > Q'_c$

T_c and deposit heat $Q_h - Qh' > 0$ into T_h. Without any external work done on the system, this transfer of heat from a cold reservoir to a hot reservoir is a direct violation of the second law of thermodynamics, meaning that our initial assumption that $\eta_E > \eta_R$ must be false. Hence, there exists no engine more efficient than a reversible engine operating between two reservoirs.

If reversible engines are the most efficient engine between to temperature reservoirs, they must have the same efficiency, which is the maximum possible efficiency. Thus, the corollary mentioned earlier is also true. In real life applications, Carnot's theorem gives the upper bound on efficiency of a given heat engine, although most commercial engines don't come close to the ideal Carnot engine ★